

Network Infrastructure Day 4 Summary

Thursday, October 8th, 2026

Overview

Key participants: QA Trainer Ninad Rampure leads the session and drives theory, examples, and lab guidance; active learners include Blake, Fabian, Simon Adams, Kai Standley, Muhammed, and others who report lab progress and ask clarifying questions.

What happened: The class covers IPv6 fundamentals and compression, NAT variants and transition mechanisms, DHCP behavior and APIPA, DNS resolution and record types, and an extended routing fundamentals segment contrasting distance-vector (RIP), hybrid (EIGRP), and link-state (OSPF) behaviors. The trainer ties each theory segment to hands-on labs (IP addressing, NAT/PAT, subnetting/VLSM, DHCP, DNS, static/RIP/EIGRP) and walks through common troubleshooting scenarios and lab image discrepancies.

Major outcomes and decisions: Students practiced prefix identification and subnetting exercises, identified preconfigured lab issues (notably a port 443 NAT entry), and agreed to proceed with at least one lab today while deferring remaining labs to the next session. Ninad recommends starting with the RIP lab for speed and plans to upload corrected notes, solutions, and any fixed lab images to GitHub.

Commitments and concerns: Ninad commits to updating lesson documents (including IPv6 content), correcting lab image preconfigurations (port-443 NAT), flagging known lab typos, and uploading materials to GitHub for download. Learners reported environment issues (interface labeling, preconfigured translations) that Ninad will help remediate.

Bottom line: The session balances protocol theory with practical lab work, ensuring learners understand addressing, NAT, DHCP, DNS, and routing choices; the class ends with a planned reconvene tomorrow to continue lab completion and deeper configuration examples.

Detailed Summary

Course progress update

The trainer reviews yesterday's progress and outlines remaining topics for the course. He explains schedule adjustments to finish routing today so the remaining objectives can be completed by Friday. The current plan prioritizes NAT, DHCP, DNS, and routing along with finishing IPv6 documentation and sharing materials on GitHub.

- QA Trainer Ninad Rampure states the team completed roughly half of Day 3 and expects to finish routing today to stay on target for Friday.
- QA Trainer Ninad Rampure says he will update the lesson document to include the IPv6 section before sharing on GitHub.
- QA Trainer Ninad Rampure indicates NAT, DHCP, DNS and routing remain as primary objectives to complete in the coming sessions.

Importance of IP addressing

The trainer emphasizes foundational IP addressing concepts (IPv4 classes, subnetting, VLSM/CIDR) and their operational role for troubleshooting, routing, and topology design. He ties these fundamentals to labs and future work, urging learners to keep the shared reference materials handy.

- QA Trainer Ninad Rampure explains that subnetting and VLSM (CIDR) are essential for network engineers to understand topology and routing.
- QA Trainer Ninad Rampure recommends learners bookmark the subnet mask and slash-notation reference he shared for quick practical lookup.
- QA Trainer Ninad Rampure notes the OSI encapsulation lab helps students inspect packet contents across layers for diagnostics.

Wireless and transport protocols

The trainer recaps wireless frequency trade-offs (2.4 vs 5GHz and newer standards) and asks students to know TCP and UDP header differences. He ties these protocol types to traffic analysis and lab exercises.

- QA Trainer Ninad Rampure highlights pros and cons of 2.4GHz and 5GHz Wi-Fi frequencies and mentions newer standards (802.11ac/ax and emerging Wi-Fi 7).
- QA Trainer Ninad Rampure reminds learners that most network traffic uses TCP or UDP and suggests inspecting headers in the command prompt or lab to analyze device communication.

IPv6 introduction and benefits

The trainer introduces IPv6 fundamentals: 128-bit address space, massive capacity, hexadecimal representation, and advantages over IPv4. He frames IPv6 as simpler to reason about (standard /64 network portion) and stresses understanding hexadecimal, compression rules, and practical host/network partitioning.

- QA Trainer Ninad Rampure states IPv6 uses 128 bits and yields an extremely large address space ($\approx 10^{38}$ addresses).
- QA Trainer Ninad Rampure explains each hexadecimal digit represents four bits and that IPv6 is typically split into eight 16-bit sections (four hex digits each).

- QA Trainer Ninad Rampure emphasizes the common /64 convention where the first four hexets are network and the last four are host.

Hexadecimal and compression rules

The trainer explains hexadecimal basics and IPv6 address compression rules: remove leading zeros, reduce groups of zeros to single zero, and use a double-colon once to compress consecutive zero hexets. He demonstrates how compressed addresses map to full addresses and why compression matters in practice.

- QA Trainer Ninad Rampure teaches that hexadecimal digits go 0–9 then A–F and each hex digit equals four binary bits.
- QA Trainer Ninad Rampure gives the three IPv6 compression rules: remove leading zeros in a hexet, shorten full-zero hexets to single zero, and use a single double-colon to compress consecutive zero hexets.

IPv6 addressing examples and types

Using public examples (Microsoft, Amazon, Facebook, Instagram), the trainer shows how IPv6 addresses reveal registry/region prefixes and how host identifiers can be chosen or reused (e.g., anycast and branded host IDs). He explains address types (global unicast, unique local, link-local, multicast, loopback) and how prefixes indicate those types.

- QA Trainer Ninad Rampure demonstrates that global IPv6 addresses typically begin with 2xxx and that regional registries (e.g., RIPE) allocate ranges indicating geography.
- QA Trainer Ninad Rampure describes anycast: multiple instances share the same host ID and clients connect to the nearest instance, using Microsoft examples showing identical host portions across regional addresses.
- QA Trainer Ninad Rampure points out examples where companies embed recognizable hex patterns (Facebook/Instagram) in host identifiers and explains uniqueness requirements and abuse risks.

IPv6 prefix identification activity

The trainer runs a live activity where learners identify IPv6 address types from prefixes. Students successfully classify examples as global, multicast, loopback, link-local, unique local, and global unicast, reinforcing the prefix-based recognition approach.

- QA Trainer Ninad Rampure asks students to identify IPv6 address types from examples and confirms correct answers during the exercise.
- Fabian identifies a 2xxx-prefixed address as global, which the trainer confirms as correct.
- Muhammed and other learners correctly identify multicast and link-local examples during the exercise.

NAT fundamentals and role

The trainer defines NAT as translation of private-to-public IP addresses at the network boundary, used to conserve public IPs and add a layer of security by masking internal topology. He explains where NAT sits (routers/firewalls) and links NAT behavior to real-world examples like load balancing and VPN considerations.

- QA Trainer Ninad Rampure explains NAT translates private IPs to public IPs and masks internal hosts from the internet.
- QA Trainer Ninad Rampure notes NAT typically runs on a router or firewall at the network perimeter and is a common reason external observers see only the router's public IP.

NAT variants and use cases

The trainer covers static NAT (one-to-one), dynamic NAT (pool-based load balancing/round-robin), PAT/port-based NAT (many-to-one by unique port), and policy-based NAT (conditional translation). He describes typical use cases for servers, load balancing, and security trade-offs.

- QA Trainer Ninad Rampure defines static NAT as a one-to-one mapping suitable for servers requiring a consistent public IP.
- QA Trainer Ninad Rampure describes dynamic NAT as using a pool of public IPs for load distribution and round-robin behavior.
- QA Trainer Ninad Rampure defines PAT (NAT overload) as mapping many private hosts to one public IP using distinct port numbers and cautions about port exhaustion in large deployments.
- QA Trainer Ninad Rampure introduces policy-based NAT to translate addresses based on conditions (destination, protocol, application) for flexible routing.

NAT advantages and limitations

The trainer lists NAT benefits (public IP conservation, security via masking, flexible integration) and drawbacks (breaks true end-to-end connectivity, additional processing latency, port limitations). He ties these trade-offs to design choices in on-prem and cloud environments.

- QA Trainer Ninad Rampure highlights NAT's advantage of reducing the need for multiple public IPs and increasing perceived security by hiding internal structure.
- QA Trainer Ninad Rampure warns that NAT can break protocols requiring end-to-end connectivity and that translation introduces processing overhead and potential latency.

IPv4-to-IPv6 transition NAT

The trainer explains NAT64/NAT46 and other transition mechanisms that allow IPv6-only clients to reach IPv4-only services and vice versa. He emphasizes practical use cases (e.g., IPv6-native devices printing to IPv4 printers) and configuration concepts for enabling the transition on routers.

- QA Trainer Ninad Rampure explains NAT64 (six-to-four tunneling) allows IPv6 clients to access IPv4 services by encapsulating and translating packets.
- QA Trainer Ninad Rampure notes transition NAT requires configuring a prefix (e.g., a 64-derived prefix) and enabling NAT64 on the router interface.
- QA Trainer Ninad Rampure states dual-direction NAT supports both inbound and outbound translation for enterprises with multiple routing domains.

Policy-based and dual NAT details

The trainer describes policy-based NAT conditions and the complexity of managing granular NAT rules, and he introduces dual NAT for bidirectional mappings between routing domains. He stresses careful rule design to avoid unintended access and administrative overhead.

- QA Trainer Ninad Rampure explains policy-based NAT lets you conditionally allow translation for specific source IPs, interfaces, or destinations for fine-grained control.
- QA Trainer Ninad Rampure warns policy NAT can be complex and mis-specified conditions may allow unintended traffic.
- QA Trainer Ninad Rampure describes dual NAT as a method to permit both private-to-public and public-to-private translations typically used in multi-campus or provider networks.

Upcoming labs and practical tasks

The trainer outlines immediate lab work: IPv4 lab, NAT lab, and optional subnetting/VLSM problems. He plans to finish remaining IPv6 theory briefly, then run back-to-back labs to reinforce concepts and will help students with remote PC access issues to ensure everyone can participate.

- QA Trainer Ninad Rampure states he will complete IPv6 theory then walk learners through IPv4 and NAT labs.
- QA Trainer Ninad Rampure offers an optional VLSM/subnetting lab for additional practice and intends to solve a subset of complex questions together.
- QA Trainer Ninad Rampure asks learners to log into the remote lab portal and offers to remotely restart services or connect to student machines to resolve access issues.

Lab 12 — IP addressing

Lab 12 requires statically configuring IPs on devices across two networks, enabling router interfaces, and verifying connectivity via ping and simulated PDUs. The trainer emphasizes common mistakes (interface left shut, wrong default gateway) and notes the first cross-network ping may fail due to ARP learning before succeeding on retry.

- QA Trainer Ninad Rampure instructs learners to statically set IPs and enable router interfaces (no shut) to ensure gateways are reachable.
- QA Trainer Ninad Rampure notes the first ping across networks may fail while ARP entries build and recommends retrying to verify connectivity.
- QA Trainer Ninad Rampure clarifies correct default gateway values for the two networks and cautions against reusing the same gateway IP on both sides.

Lab 13 — NAT and PAT

Lab 13 tests configuring NAT and PAT so hosts in the lab network can access servers mapped through public IPs; HTTP should succeed while HTTPS may be intentionally blocked by PAT. The trainer provides lab verification steps and flags a preconfigured issue in the lab image that may cause HTTPS to pass unexpectedly.

- QA Trainer Ninad Rampure explains the lab maps public IPs to private servers and verifies HTTP/HTTPS behavior through NAT/PAT rules.
- QA Trainer Ninad Rampure alerts learners that PAT in this lab is intentionally restricted to port 80 so HTTPS should fail when correctly configured.
- QA Trainer Ninad Rampure acknowledges a preconfiguration issue reported by Blake where port 443 was already present in the lab image and commits to flagging and clearing the image if needed.

Lab troubleshooting and environment notes

Learners report discrepancies in the lab images (preconfigured NAT entries, interface labeling differences); the trainer inspects examples, confirms misconfigurations, and offers immediate remediation steps or follow-ups. He encourages students to report anomalies while proceeding with remaining lab tasks.

- Blake reports and Ninad confirms a preconfigured port-443 translation exists on the lab image, which can mask the intended PAT behavior.
- Fabian reports interface numbering display differences between CLI and GUI, and Ninad clarifies how serial interface labels appear in the simulator.
- QA Trainer Ninad Rampure offers to reset or remove unintended configuration lines from the lab image and to notify other trainers if the shared image requires correction.

Subnetting activity (Lab 14 & VLSM)

The trainer walks learners through subnetting exercises: converting networks into required subnets, calculating borrowed bits, usable hosts, new masks, and identifying network and broadcast addresses. He also covers VLSM planning (Lab 15) mapping variable host requirements to appropriate slash notations and using /30 links between routers.

- QA Trainer Ninad Rampure demonstrates using the formula 2^n for subnets and $2^{(\text{host bits})} - 2$ for usable hosts and applies it to /25 and /26 examples.
- QA Trainer Ninad Rampure guides learners through VLSM: selecting /26 for ~50 hosts, /23 for ~300 hosts, and /21 for ~1500–2000 hosts, and recommends /30 for point-to-point router links.
- QA Trainer Ninad Rampure recommends using the shared subnet calculator website and the GitHub notes to verify subnet ranges, first/last IP, and broadcast addresses.

Administrative follow-ups and next agenda

The trainer commits to uploading solution documents and updated notes to GitHub, to investigate and correct lab image preconfigurations, and to resume with DHCP, DNS, and routing after the lunch break. He schedules a short break and reconvenes to start hands-on lab work.

- QA Trainer Ninad Rampure commits to uploading IP addressing and activity solution documents to GitHub for later review.
- QA Trainer Ninad Rampure will check and, if necessary, clear preconfigured NAT entries in the lab images and inform other trainers if image fixes are required.
- QA Trainer Ninad Rampure plans to continue with DHCP, DNS, and routing after the break and to spend the next hour focused on completing labs 12 and 13.

Lab 16 & 17 availability

The trainer notifies learners that lab 16 files may be missing from the provided image and advises students to design the topology in Packet Tracer using the provided IP table when necessary. He confirms lab 17 can reuse the same topology and recommends completing both labs together for efficiency.

- QA Trainer Ninad Rampure acknowledges lab 16 may be missing from the simulator image and instructs students to design the topology in Packet Tracer using the IP table if required.
- QA Trainer Ninad Rampure advises learners to reuse the same network design for lab 17 to avoid rebuilding the topology.

- QA Trainer Ninad Rampure reminds students to select appropriate device models (ISR routers, 2960 switches) and match interfaces (router G0/1 for inter-router links, switch Gi ports for hosts) when creating the topology.

DHCP — concepts and process

The trainer explains DHCP as the dynamic IP management service, covering its discover-offer-request-acknowledge (DORA) exchange, lease behavior, renewal windows (T1/T2), and fallback APIPA addressing when no server responds. He stresses lease sizing trade-offs to balance IP pool exhaustion and broadcast storm risk.

- QA Trainer Ninad Rampure defines DHCP as Dynamic Host Configuration Protocol and describes it as an IP address management service that issues leases.
- QA Trainer Ninad Rampure explains the DORA process (Discover, Offer, Request, Acknowledge) and that DHCP uses UDP ports 67 (server) and 68 (client).
- QA Trainer Ninad Rampure highlights lease renewal timing (T1 ~50% and T2 ~87.5%) and notes that expired leases trigger a full DORA cycle which increases broadcast traffic.
- QA Trainer Ninad Rampure warns that inappropriate lease lengths can exhaust IP pools for occasional visitors or create renewal broadcast bursts for many clients simultaneously.

APIPA and DHCP failure handling

The trainer covers APIPA behavior when DHCP is unavailable and how clients self-assign 169.254.x.x addresses, and he shows lab exercises to simulate DHCP failure and recovery. He explains how release/renew interactions restore proper addressing once the DHCP service returns.

- QA Trainer Ninad Rampure explains APIPA assigns 169.254.x.x addresses when a DHCP server does not respond.
- QA Trainer Ninad Rampure states the only remedy for APIPA is to restore the DHCP server and run release/renew on clients to obtain addresses from the server.

DNS — role and resolution

The trainer defines DNS as name-to-IP resolution service, distinguishes it from ARP and routing tables, and explains forward and reverse lookups. He walks through recursive resolution: client cache → local DNS → forwarder → root → TLD → authoritative domain, and explains caching and TTL implications.

- QA Trainer Ninad Rampure clarifies DNS maps human-readable names to IPs (forward lookup) and can perform reverse lookup via PTR records.

- QA Trainer Ninad Rampure outlines recursive resolution from client cache, to local DNS/forwarder, to root servers, then TLD servers, and finally the authoritative server for the domain.
- QA Trainer Ninad Rampure explains DNS responses are cached with TTL values so subsequent queries can be answered locally until the TTL expires.

DNS record types and zones

The trainer reviews common DNS resource records (NS, SOA, A, AAAA, PTR, CNAME, MX) and explains zones as collections of records served by authoritative servers. He emphasizes how records support services (web, mail, aliases) and how zones/local SOA assignment affects resolution behavior.

- QA Trainer Ninad Rampure identifies A records as name→IPv4, AAAA as name→IPv6, and PTR as the reverse IP→name record.
- QA Trainer Ninad Rampure explains CNAME records create aliases (e.g., www → actual A record) and MX records point mail delivery to mail servers.
- QA Trainer Ninad Rampure describes SOA records as indicating the authoritative DNS server for a zone and NS records as listing other name servers for redundancy.

DNS lab instructions and challenge

The trainer assigns labs 16 and 17 (DHCP and DNS) to be completed together, clarifies design/configuration steps for Packet Tracer, and issues a challenge activity requiring a custom topology and correct DHCP subnet mask selection. He encourages finishing core labs before attempting the challenge due to time constraints.

- QA Trainer Ninad Rampure instructs learners to design the Packet Tracer topology (routers, switches, servers) if the lab image is missing and then proceed to configuration steps.
- QA Trainer Ninad Rampure asks students to ensure DHCP server and router IPs use the correct subnet (challenge requires /28) and to enable interfaces and DHCP service during configuration.
- QA Trainer Ninad Rampure recommends students finish labs 16 and 17 first and attempt the extension challenge only if time permits.

Routing concepts introduction

The trainer begins layer-3 routing instruction, framing routing as essential for end-to-end communication and previewing protocol types (interior vs exterior). He asks knowledge-check questions and outlines the upcoming deeper dive into OSPF, EIGRP, and BGP differences and use cases.

- QA Trainer Ninad Rampure frames routing as layer-3 responsibility for moving traffic across multiple routers to reach destinations and previews protocol comparisons (OSPF, EIGRP, BGP).
- QA Trainer Ninad Rampure asks learners to assess prior knowledge of routing protocols to tailor the depth of the upcoming discussion.

Routing protocol types overview

The trainer contrasts interior gateway protocols (IGP) like OSPF and EIGRP — used inside an autonomous system — with exterior gateway protocol (BGP) used between autonomous systems. He introduces key concepts like link-state vs distance-vector behavior and the role of autonomous systems in routing design.

- QA Trainer Ninad Rampure explains OSPF and EIGRP are IGPs used within an autonomous system and that BGP is an EGP used between autonomous systems (e.g., ISPs).
- Fabian observes that OSPF uses link-state advertisements to update topology and the trainer affirms OSPF's Open Shortest Path First nature.
- QA Trainer Ninad Rampure introduces the core distinction between link-state and distance-vector metrics and how shortest-path calculations influence route selection.
- QA Trainer Ninad Rampure states that RIP is simple and easy to deploy but limited to 15 hops, making it unsuitable for large or complex networks.
- QA Trainer Ninad Rampure explains EIGRP is a hybrid protocol that considers link characteristics like bandwidth and delay to select routes.

Routing protocol selection factors

The trainer leads a discussion on how topology, scalability, performance, and management needs determine routing protocol choice. He uses example topologies (linear vs hierarchical) to illustrate why RIP, EIGRP, OSPF, or BGP may be preferable in different scenarios and stresses correct deployment and convergence planning.

- QA Trainer Ninad Rampure states that network topology and design are primary factors in choosing a routing protocol for an enterprise.
- QA Trainer Ninad Rampure explains RIP suits simple, linear topologies due to ease of deployment but is limited by a 15-hop maximum.
- QA Trainer Ninad Rampure advises OSPF for hierarchical, enterprise-scale networks because it minimizes regular updates and only floods changes, reducing resource use.
- QA Trainer Ninad Rampure explains BGP is appropriate for ISP-level edge routing between autonomous systems because providers do not need to know internal topology.

Link-state vs distance-vector behaviors

The trainer clarifies how link-state protocols evaluate link metrics and propagate topology changes versus distance-vector protocols that rely on route distance (hops) and neighbor exchanges. He contrasts EIGRP as a hybrid that factors bandwidth/delay with pure distance or link-state approaches, and he uses analogies to make convergence behavior tangible.

- QA Trainer Ninad Rampure explains link-state protocols compute shortest paths using link metrics and propagate updates primarily when topology changes occur.
- QA Trainer Ninad Rampure describes distance-vector protocols as selecting routes based on distance metrics such as hop count and periodically sharing full routing information with neighbors.
- QA Trainer Ninad Rampure explains EIGRP behaves as a hybrid by using multiple link characteristics (bandwidth, delay) to calculate best paths and improve routing decisions.
- QA Trainer Ninad Rampure notes distance-vector protocols (like RIP) send frequent updates (e.g., every 30 seconds) which can consume bandwidth and delay convergence for large networks.
- QA Trainer Ninad Rampure highlights link-state protocols (like OSPF/IS-IS) converge faster because routers maintain a common topology view and only flood incremental changes.

Distance-vector specifics and RIP

The trainer details RIP behavior: hop-count metric, neighbor-based learning, version differences, classless support in RIP v2, limitations including 15-hop maximum, susceptibility to routing loops, and slower convergence compared with link-state protocols. He recommends RIP v2 only for very small or simple topologies and points learners to routing notes on GitHub for visual examples.

- QA Trainer Ninad Rampure explains RIP calculates best path by hop count and only knows routes advertised by immediate neighbors.
- QA Trainer Ninad Rampure recommends RIP version 2 over version 1 because RIP v2 supports classless addressing (VLSM/CIDR) and offers basic authentication.
- QA Trainer Ninad Rampure warns that RIP's 15-hop limit and slower, periodic updates make it unsuitable for large or complex network designs.

Convergence and routing loops

The trainer explains convergence behavior differences across protocols and demonstrates how routing loops form when default routes or misconfigurations cause routers to forward queries in circles. He contrasts router behavior (packets with TTL) with switch behavior (frames without TTL) and introduces STP as the mitigation for switching loops.

- QA Trainer Ninad Rampure states link-state protocols converge quickly on events because each router has a full topology and only floods changes.
- QA Trainer Ninad Rampure warns that misconfigured default routes can create routing loops that repeatedly forward packets between routers until TTL expires.
- QA Trainer Ninad Rampure explains switches can create persistent broadcast loops because frames lack TTL and recommends Spanning Tree Protocol (STP) to block redundant switch ports.

Static routes — use and tradeoffs

The trainer outlines when static routes are appropriate: small, stable networks with limited routers and resource constraints. He explains static routes reduce router processing and can improve security by preventing untrusted devices from injecting routes, but they require manual updates and risk human error in larger deployments.

- QA Trainer Ninad Rampure explains static routes suit small, stable topologies because they require no routing protocol overhead and conserve router resources.
- QA Trainer Ninad Rampure notes static routing improves control and can limit unexpected route propagation, which can be beneficial for security.
- Simon Adams and Fabian highlight that static routing increases documentation and manual maintenance burden and is error-prone as the network scales.

Dynamic routing tradeoffs

The trainer contrasts dynamic routing's adaptability with the control cost: dynamic protocols automate route learning and reduce manual work but can lose administrator control, consume bandwidth during updates, and exhibit delayed propagation depending on the protocol. He emphasizes choosing a protocol that matches network scale and performance goals.

- QA Trainer Ninad Rampure explains dynamic routing automates route distribution and reduces manual configuration effort across many routers.
- QA Trainer Ninad Rampure warns that dynamic routing reduces direct control because routers autonomously share and update routes, which can propagate misconfigurations if not managed.
- QA Trainer Ninad Rampure highlights resource tradeoffs: dynamic routing consumes CPU, memory, and bandwidth to compute and exchange route information.

Routing labs and practical guidance

The trainer assigns static, RIP, and EIGRP labs and recommends starting with the RIP lab for a quick hands-on exercise, then completing static and EIGRP labs later. He

explains lab steps (enable RIP v2, advertise networks, use show ip route) and flags a minor typo in one lab step that students should watch for.

- QA Trainer Ninad Rampure recommends starting with the RIP lab because it is the fastest to complete, then doing static and EIGRP labs tomorrow.
- QA Trainer Ninad Rampure instructs learners to configure routers for RIP version 2, advertise directly connected networks, and verify convergence using show ip route.
- QA Trainer Ninad Rampure alerts students to a known typo in step four of the RIP lab and asks them to follow the corrected step five example.

Routing table indicators

The trainer reviews routing table notation so learners can identify how entries were learned (connected, static, or via a protocol). He explains common symbols used by routers and how to interpret them during lab verification.

- QA Trainer Ninad Rampure explains routing table entries show a code: 'C' for directly connected, 'S' for static, 'R' for RIP, 'E' for EIGRP, and 'O' for OSPF.
- QA Trainer Ninad Rampure advises learners to use show ip route to inspect how routes were learned and to validate lab configurations.

Lab troubleshooting and wrap-up

Learners report progress and issues on static and RIP tasks; the trainer offers guidance, encourages finishing one lab today, and schedules remaining labs for the next session. He reminds students where to find updated lesson summaries and GitHub files and requests learners download the updated materials.

- QA Trainer Ninad Rampure advises students to complete at least one lab today and continue static and EIGRP labs tomorrow morning.
- Blake confirms he has finished a lab and Ninad permits early finishers to leave for the day.
- QA Trainer Ninad Rampure reminds learners he will upload lesson summaries and updated files to GitHub and asks students to download the latest folder updates.
- QA Trainer Ninad Rampure ends the session and confirms the class will reconvene tomorrow for remaining work.